

Chinmay Appa Rane

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SUMMARY

Ph.D. ML Engineer with 7+ years of applied research and industry experience building multimodal and medical imaging AI systems, including self-supervised representation learning and clinical deployments across hospitals. ICLR-published researcher and NVIDIA GTC speaker.

TECHNICAL SKILLS

Languages	Python, C++, MATLAB, SQL
ML/AI Frameworks	PyTorch, TensorFlow, Keras, MONAI, Hugging Face Transformers, Scikit-learn, DeepSpeed, LLaMA Factory, Unsloth, LangChain, CrewAI, LangGraph
AI/ML Techniques	Vision (Segmentation, Object Detection), LLMs, OCR, Multimodal VLMs, RAG, Self-Supervised Learning, Federated Learning, Transformers, Diffusion Models, PEFT, Full Supervised Fine-Tuning
Production & Deployment	TensorRT, ONNX, ONNX Graph Surgeon, vLLM, TensorRT-LLM, Triton Inference Server, Docker, Docker Compose, Kubernetes, FastAPI, REST APIs
Cloud & MLOps	AWS (SageMaker, EC2, S3), GCP, CI/CD, MLflow, Databricks, Milvus Vector DB, Prometheus, Grafana
Domains	Medical Imaging, Autonomous Systems, Generative AI, Technical Leadership

WORK EXPERIENCE

Associate Technical Architect - Quantiphi Inc

(September 2021- Current)

Leading multimodal and generative AI initiatives across LLMs, VLMs, and medical imaging, mentoring teams and delivering production systems from research to real-world deployment.

- **Healthcare Revenue Cycle Denial Resolution System** - Led architecture and development of intelligent denial resolution platform leveraging historical claims data and human intervention patterns. Designed embedding-based retrieval system (K-means + cosine similarity), engineered claim context generation pipeline, and integrated LLM reasoning layer. Drove technical decisions for dual-mode deployment (pre-submission prevention, post-payer resolution), currently in production.
- **Clinical EHR LLM Model Building and Optimization** - Engineered DeepSpeed/Unsloth training pipeline reducing LLaMA fine-tuning time by 55% (42→19 hours, \$15K+ savings/cycle) across 8x H100 GPUs (95%+ efficiency). Implemented vLLM inference for production scale while mentoring team on distributed training.
- **Enterprise-grade AI platform for cardiovascular diagnostics** - Deployed self-supervised MONAI pipeline reducing annotation costs 10× (\$50K→\$5K) and report generation time by 50% with 40% accuracy improvement. Production system on AWS/Triton serving 7 hospitals at 85% accuracy for surgical planning.
- **Structural Repair Document Creation using VLMs & LLMs** - Built VLM/LLM system (InternVLM, LLaMA 3.2, RAG) reducing helicopter ground time by 70% (\$50K+ daily savings) through automated damage detection and repair report generation. Led technical implementation and customer relations.
- **3D Texture Generation using Vision-Language** - Developed VLM-based automation reducing texturing time by 80% (40→8 hours) using CLIP embeddings with DBSCAN clustering for zero-shot component recognition. Co-led dual-deployment architecture (laptop/server) for manufacturing workflows.

ADDITIONAL WORK EXPERIENCE

Neural Network Research Assistant - Image Processing and Neural Networks Lab, UTA

(2017- 2021)

Research on neural networks for medical imaging (LASIK) and geophysical applications in C++, MATLAB, Python.

Data Scientist Intern - Unique Software Development

(Jan 2018 – Dec 2018)

Built production NLP pipelines and object detection systems for transportation analytics.

EDUCATION

University of Texas at Arlington | Doctor of Philosophy, *Deep Neural Networks* | Electrical Engineering

2021

University of Texas at Arlington | Master of Science, *Neural Networks* | Electrical Engineering

2016

PUBLICATIONS AND THOUGHT LEADERSHIP

- Chinmay Rane, Michael Manry, "Dynamic Activations for Neural Net Training", ICLR 2024
- NVIDIA GTC 2025 invited speaker - "Revolutionizing Cardiac MRI Analysis and Diagnosis With AI"
- 3 peer-reviewed publications on neural network architectures and training (ArXiv, SN Computer Science)